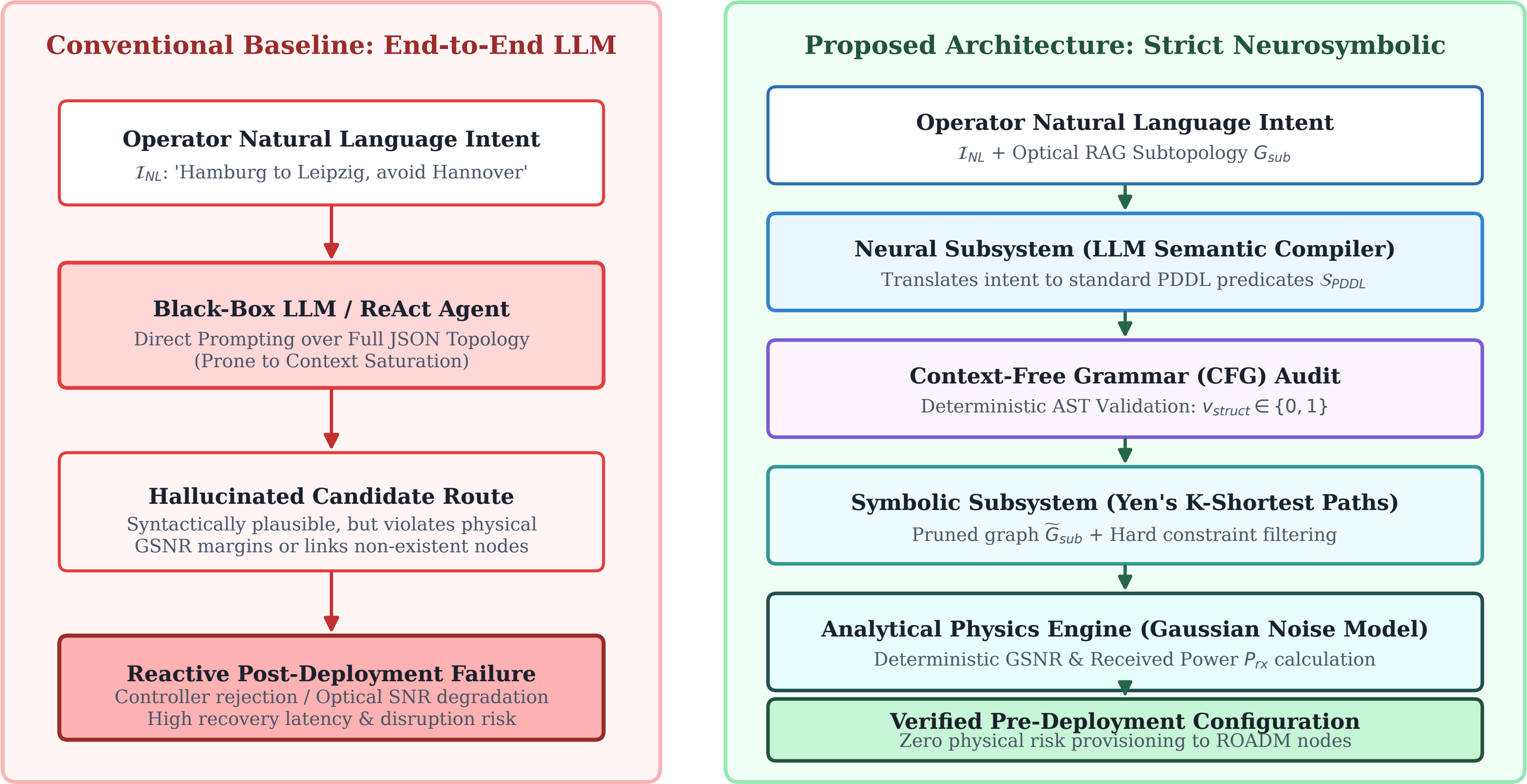
Conventional Baseline: End-to-End LLM
Operator Natural Language Intent
$\mathcal{I}_{NL}$: 'Hamburg to Leipzig, avoid Hannover'
Black-Box LLM / ReAct Agent
Direct Prompting over Full JSON Topology
(Prone to Context Saturation)
Hallucinated Candidate Route
Syntactically plausible, but violates physical
GSNR margins or links non-existent nodes
Reactive Post-Deployment Failure
Controller rejection / Optical SNR degradation
High recovery latency & disruption risk
Proposed Architecture: Strict Neurosymbolic
Operator Natural Language Intent
$\mathcal{I}_{NL}$ + Optical RAG Subtopology G_{sub}
Neural Subsystem (LLM Semantic Compiler)
Translates intent to standard PDDL predicates $\mathcal{S}_{PDDL}$
Context-Free Grammar (CFG) Audit
Deterministic AST Validation: $v_{struct} \in \{0, 1\}$
Symbolic Subsystem (Yen's K-Shortest Paths)
Pruned graph $\widetilde{G}_{sub}$ + Hard constraint filtering
Analytical Physics Engine (Gaussian Noise Model)
Deterministic GSNR & Received Power P_{rx} calculation
Verified Pre-Deployment Configuration
Zero physical risk provisioning to ROADM nodes